

Aerospace Performance Toolkit

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The purpose of this project was to build a program capable of computing and graphing various aerospace equations and their properties. Focusing on standard atmosphere, compressible flow, aerodynamics and orbital mechanics calculations. The properties of the equations used were also graphed, some with the inclusion of data inputs so they can be used for specific purposes.

For the first part of the menu, calculations for standard atmosphere were performed. Temperature, pressure, density and speed of sound were calculated using altitude as the independent variable. Due to time constraints, the calculations are only accurate for the first 11 km. This is because the equations defer depending on altitude and the first section of atmosphere was calculated. After the calculations the user then can produce graphs for each of the four variables mentioned above up to the constraint. This allows the user to be able to see the calculations and have a visual of the effects of altitude on the variable.

The second section deals with the isentropic flow relationships as a function of the Mach number. This includes temperature ratio, pressure ratio, density ratio, stagnation temperature, stagnation pressure, area ratio and Mach angle. Using the Mach number is relevant to the aerospace related calculations and graphs and allows for the flow behavior to be seen easier. The other method would be to use the equations at one specific operating condition. This is often seen in propulsion and engines. The pressure, temperature and area ratios were then graphed using the Mach number as the independent variable. These ratios are critical in engine performance calculations that were used in the rocket propulsion suit project. This method is fast to make the calculations with the given input and deliver results and imagery to further understanding and make predictions.

The Aerodynamics section contains many important calculations for various properties. The focus was on dynamic pressure, lift, drag, Reynolds number and the lift to drag ratio. Dynamic pressure is the main component in calculating lift and drag, but it is still important on its own. It is a factor of velocity and air density which was calculated based on altitude. It has major impacts on the calculations alongside the lift and drag coefficients. The viscosity is one of the most important details in aerodynamics and can showcase air as a fluid like behavior. It is great at measuring the resistance of a flow and is often used to determine how much turbulence is produced by a plane's wing. The lift, drag and the ratio were also able to be plotted as a variable of velocity. While the lift and drag are different values due to their

confidences, the ratio between them stays the same at constant conditions since velocity impacts both equally. This section helps to expand the interface to cover a wider view of the aerospace sector.

The final finished section focuses on orbital mechanics calculations based on input altitudes. The circular orbital velocity, escape velocity, orbital period, and the Hohmann transfer velocity change are calculated. The first two velocities used the Earth's radius to calculate the correct gravitational pull on the object with its altitude. The period calculation also used a circular orbit. The more complex calculation is the Hohmann transfer which calculates the change in velocities for orbits. Changing most orbits requires two different burns, the first one is elliptic and the second one a pi radian later to even it out back to a circle. Both the velocities and total are calculated and displayed. The plots generated compare circular orbital velocity and orbital period to altitude. This shows that a higher altitude produces a longer period while velocity stays relatively the same.

While a wide range of topics is covered throughout the toolkit it can be expanded to cover other sectors. The current sections can also be expanded on with more advanced calculations that take more variables. This would create more aerodynamics performance and orbital mechanics calculations. More advanced models like the atmosphere, supersonic and hypersonic flow could be made. They would help to create a better understanding through advanced visuals. With more allowed to complete this project that features would have been added in creating more depth. The toolkit is successful at calculating fundamental aerospace equations and creating plots. It has many useful features that can be used in the day to day operations of an engineer and is able to link together various aspects of aerospace engineering.